

Course Code: CSE316/3083	Course Title: Advanced Computer Architecture	L-T-P-C	3-0	0	3
	Type of Course: Program Core & Theory Only				
Version No.	1.0				
Course Pre-requisites	NIL				
Anti-requisites	NIL				
Course Description	The course aims at familiarizing students with advanced computer architectures suitable for high-performance computing. The advanced concepts in uniprocessor and the issues in designing & using high performance parallel computers will also be covered. System resources such as memory technology and I/O subsystems needed to achieve proportional increase in performance will be discussed along with the software support required for these systems.				
Course Objective	The objective of the course is to familiarize the learners with the concepts of Advanced Computer Architecture and attain Employability through Participative Learning techniques .				
Course Out Comes	On successful completion of the course the students shall be able to: 1] Explain the concepts of parallel computing and hardware technologies 2] Compare and contrast the parallel architectures 3] Illustrate parallel programming concepts 4] Understand the organization and operation of current generation parallel computer systems, including multiprocessor and multicore systems.				
Course Content:					
Module 1	Theory of Parallelism	Assignment			10 Sessions
Topics: Theory of Parallelism: Parallel Computer Models, The State of Computing, Multiprocessors and Multicomputer, Multivector and SIMD Computers, PRAM and VLSI Models, Program and Network Properties, Conditions of Parallelism, Program Partitioning and Scheduling, Program Flow Mechanisms, System Interconnect Architectures, Principles of Scalable Performance, Performance Metrics and Measures, Parallel Processing Applications, Speedup Performance Laws, Scalability Analysis and Approaches.					